Kundai Farai Sachikonye

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Microscopy image analysis and computer vision, with machine-learning representation learning for cell modelling — and the biology to ground it.

Profile

Computational researcher in **microscopy image analysis and computer vision**, with hands-on **machine-learning** and a background in **cell biology and bioinformatics**. I have built and deployed a quantitative microscopy image-analysis framework, and I am keen to formalise this work into a PhD on large-scale 3D microscopy analysis, shape modelling, and representation learning for cells.

Selected Work

Microscopy image analysis: **Quantitative image-analysis framework** — segmentation, point-spread-function deconvolution, scale-field / coordinate (geometry) reconstruction, and morphometry with uncertainty quantification, for fluorescence and phase-contrast microscopy. GPU-accelerated (WebGL/GLSL); live, deployed tool. hieronymus-omega.vercel.app

Shape & geometry modelling: **Coordinate-field reconstruction** — recovering metric-preserving geometry from image observations (scale fields, geodesic distance), the basis for quantitative 3D shape characterisation of biological structures.

ML / representation learning: Deep learning in **PyTorch** (CNNs, autoencoders / representation learning, attention); foundation-model fine-tuning (LoRA); applied to spectral and imaging data and to data-driven cell/disease-state modelling.

Cell modelling: **Cell-state characterisation** — spectral and circuit models of cellular state for diagnostics (tumour disease state, immunopeptidomics), connecting imaging and molecular data.

Experience

2021–present **Independent Computational Researcher**, *Bitspark GmbH*
Full-time research in microscopy image analysis, computer vision, machine learning, and computational biology. Built and deployed open, documented, GPU-accelerated tools; authored scientific papers.

2019–2021 **Computational Lipidomics — Bioinformatics & HPC**, *Technische Universität München*, Munich
Large-scale high-throughput data analysis at HPC scale; CNN-based classification (PyTorch); statistical evaluation of >100 GB datasets. Taught at Master's level and supervised student work.

2017–2018 **Glycomics and Proteomics**, *Universitätsklinikum Hamburg-Eppendorf*, Hamburg
Automated high-throughput LC-MS/MS and MALDI-TOF data pipelines.

Education

2018–2019 **MSc Computational Biology**, *Universität Tübingen*
Image and sequence analysis, statistics, and machine learning for biological data.

2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
Molecular and cell biology, analytical methods.

2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Thesis: DNA interstrand crosslinking — cell culture and fluorescence microscopy.

Additional: doctoral research in computational lipidomics at TUM (2019–2021; not completed).

Technical Skills

Image analysis / CV	Segmentation, deconvolution, registration, morphometry; coordinate-field / geometry reconstruction; GPU/GLSL image processing; 2D and (towards) 3D
Machine learning	PyTorch (CNNs, autoencoders, attention); representation learning; foundation-model fine-tuning (LoRA); scikit-learn
Programming	Python (primary), Rust, TypeScript/WebGL, R, Bash, SQL
Biology	Cell biology, biochemistry, bioinformatics; transcriptomics, proteomics, metabolomics

Languages

English Native / Fluent (C2)

German Conversational *resident in Germany since 2011*